SBMS REQUIREMENTS AND SCOPE CATALOGUE

Prepared: 2026-11-02 **System:** Student Beneficiary Management System (SBMS) **Revision described:** `fix/sbms-completion-work` @ `d6b5032` **Evidence basis:** source, migrations, tests and recorded run documents in the repository. Nothing in this catalogue is a plan; every row describes what exists at the revision above. **Companion files:** `SBMS_API_ENDPOINT_CATALOGUE.md`, `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md`, `SBMS_TEST_EVIDENCE_CATALOGUE.md`, `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md`.

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1. Purpose

This file is the single requirements register for the written report. It exists so that Chapters Four, Five and Six can cite one authoritative numbered list instead of restating requirements in prose that drifts between chapters. It states, for each requirement: the identifier, the requirement as a testable statement, the actor, the priority, the implementation location, the evidence that establishes the behaviour, and the verification status.

The identifiers are those already used in `CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` §4.2.2 (FR-01 … FR-37) and §4.2.3 (NFR-01 … NFR-21). They are reproduced here, not renumbered, so cross-references in the chapters remain valid.

2. Status vocabulary

Code	Meaning	Test applied
V	Verified	An executed automated test or a recorded observation establishes the behaviour at this revision
P	Partially verified	Some behaviour is established; a named part is not (the row says which)
N	Implemented, not tested	Code path exists and was read; no executed test exercises it
X	Not implemented	Absent by scope decision or recorded as a gap

A requirement is called *implemented* only where a route, a service, a database object and a user-facing affordance all exist. Where the interface exists but no executed test covers the behaviour, the row is marked **N**, never **V**.

3. System identity and component inventory

Layer	Technology at this revision	Entry point
Presentation	React 18 single-page application, Vite build, `react-router-dom`	`frontend/src/main.jsx`, `frontend/src/App.jsx`
Application	Node.js 22 + Express 4 REST API, 13 routers	`backend/src/index.js`
Domain services	10 service modules (screening, aggregation, review policy, eligibility, bursary rules, certificates, audit, readiness)	`backend/src/services/`
Persistence	PostgreSQL 17, 13 tables, 14 forward migrations + 1 rollback script	`backend/src/db/schema.sql`, `backend/src/db/migrations/`
Machine learning	Python 3.11, scikit-learn + XGBoost 3.2.0, one six-feature model, one long-running scoring subprocess	`ml-service/predict.py`, `ml-service/feature_engineering.py`
Delivery	Single container image (API + ML runtime), GitHub Actions build, Azure Container Apps	`Dockerfile`, `.github/workflows/publish-api-image.yml`

4. Actors and interests

Actor	Description	Where they act
Field Officer	Registers beneficiaries at a school, screens them, records verification evidence, submits award applications, records payments	`/students`, `/students/register`, `/eligibility`, `/disbursements`, `/field-officer`

Actor	Description	Where they act
Administrator	Adjudicates duplicate cases, approves or rejects awards, issues and revokes certificates, manages users and cut-offs, reads the audit log and reports	`/dashboard`, `/duplicate-review`, `/disbursements`, `/reports`, `/settings`, `/audit`, `/users`
Public verifier	Student, guardian, school or employer holding a printed QR code	`/verify`, `/verify-certificate` (unauthenticated)
Operator / deployer	Applies migrations, sets secrets, starts and stops the deployment, restores a backup	`backend/scripts/migrate.js`, `docs/deployment.md`, `deploy/azure/**`
Programme owner	Owns the open policy decisions (cut-off choice, adjudication protocol, acknowledgement deadline)	Not represented in the interface; see `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §6

5. Scope boundary

In scope and delivered: authentication and role separation; beneficiary registration with a five-school constraint; six-feature ML-assisted duplicate screening with human adjudication; rule-based eligibility scoring; bursary award submission, verification, approval and rejection; payment issue, clearance and officer-assisted receipt acknowledgement with exception reporting; QR student identification (existence only); QR-verifiable award certificates with revocation; dashboard, PDF export, audit log, cut-off settings and user administration; migrations, readiness guards and a containerised deployment.

Delivered beyond the approved proposal: award verification evidence, payment reconciliation (issue/clearance/acknowledgement with discrepancy handling), and award certificates. Recorded as expansion, not as a proposal obligation.

Excluded and stated as excluded: NEMIS or any national identifier verification service (FR-32); offline operation and native mobile applications (FR-33); financial transfer execution (FR-37 — SBMS records payments, it does not move money); multi-tenancy; token expiry and re-issue for student QR codes (FR-35, FR-36).

6. Functional requirements register (FR-01 … FR-30)

Priority: **M** must have, **S** should have, **C** could have. Verification names the file in `backend/tests/` unless a full path is given.

ID	Requirement (abbreviated statement)	Actor	Pri	Implementation	Verification	Status
FR-01	Authenticate against a stored bcrypt hash and issue a JWT valid for 8 hours	Admin, Officer	M	`auth.controller.js`	`auth.test.js`, `roleAuthorization.test.js`	V
FR-02	Separate administrator and field-officer sign-in; refuse the wrong endpoint with 403	Both	M	`auth.controller.js:44-60`	`auth.test.js`, `roleAuthorization.test.js`	V
FR-03	Refuse public self-registration (410); accounts created only by an administrator	Admin	M	`auth.routes.js:9`	`auth.test.js`	V
FR-04	Enforce the role matrix on every endpoint; 403 otherwise	All	M	`middleware/requireRole.js`; every router	`roleAuthorization.test.js` (6), `workflow.test.js`	V
FR-05	Register a beneficiary with validated name, DOB, school, national ID, phone and optional UPI / birth certificate	Officer	M	`students.controller.js:15-24`	`students.test.js` (9)	V
FR-06	Reject a school outside the five approved schools	Officer	M	`config/schools.js`; `students_school_supported_check`	`students.test.js`	V

ID	Requirement (abbreviated statement)	Actor	Pri	Implementation	Verification	Status
FR-07	Score a new registration against every existing record and raise review cases per policy	System	M	`students.controller.js:69-124`; `services/duplicateCheck.js`	`duplicateFlags.test.js`, `workflow.test.js`, `mlPredictor.integration.test.js`	V
FR-08	Never reject, delete or amend an existing record because of screening; the flagged student is still created	System	M	`students.controller.js:101`	`workflow.test.js` #1; `docs/business-workflow-verification-2026-10-30.md` §4	V
FR-09	Store score, model version, review threshold and feature vector per case	System, Admin	M	`students.controller.js:103-109`; migration `20261022_ml_duplicate_provenance.sql`	`duplicateFlags.test.js`	V
FR-10	Mask identifiers in the queue; reveal only on an explicit audited request	Admin	M	`duplicateFlags.controller.js:37-42,106-107`	`duplicateFlags.test.js`	V
FR-11	Record a reviewer decision with a mandatory reason, appended to case history	Admin	M	`duplicateFlags.controller.js:452-503`; `duplicate_review_events`	`duplicateFlags.test.js`, `workflow.test.js`	V
FR-12	Compute eligibility from five weighted factors and assign one of four categories	Officer	S	`services/eligibility.js`; `eligibility.controller.js`	Route and validation only; no automated test of the arithmetic	P
FR-13	Submit a bursary award application for an existing student with a requested amount	Officer	S	`bursaryAwards.controller.js:64-101`	`bursaryAwards.controller.test.js` (5)	V
FR-14	Record a verification outcome with an evidence reference; `verified` requires one	Officer	S	`bursaryAwards.controller.js:104-148`	`bursaryAwards.controller.test.js`; `workflow.test.js` #2	V
FR-15	Approve or reject with a reason; approval refused unless verified and no open or confirmed duplicate exists	Admin	M	`bursaryAwards.controller.js:151-209`	`bursaryAwards.controller.test.js`; `workflow.test.js` #2-#6	V
FR-16	Record payment as `issued`; refuse unapproved, unverified, duplicate-blocked or over-amount payments	Admin, Officer	M	`disbursements.controller.js:65-136`	`disbursements.test.js` (4); `workflow.test.js` #7	V
FR-17	Record clearance separately from issue, with a mandatory evidence reference	Admin, Officer	S	`disbursements.controller.js:138-191`	`disbursementReconciliation.controller.test.js` (4); `workflow.test.js` #8	V
FR-18	Record officer-assisted acknowledgement; classify a shortfall as `discrepancy` without overwriting the recorded amount	Admin, Officer	S	`disbursements.controller.js:193-260`; `bursaryRules.receiptStatus`	`workflow.test.js` #9; `bursaryRules.test.js` (4)	V
FR-19	Report reconciliation exceptions (unlinked legacy, unverified, partial, awaiting, overdue, failed, cancelled, discrepancy)	Admin, Officer	S	`bursaryRules.exceptionReasons`; `disbursements.controller.js:53-63`	`workflow.test.js` covers exercised paths; not every reason code is individually asserted	P
FR-20	Generate a cryptographically random QR token per student and return a QR image	System, Officer	M	`students.controller.js:58-60`; `config/publicAppUrl.js`	`verification.test.js` (10)	V
FR-21	Public endpoint reports only whether a matching record exists	Public	M	`verification.controller.js`; `index.js:65`	`verification.test.js`; `verificationRoute.test.js` (4)	V
FR-22	Issue a QR-verifiable certificate for an approved award; refuse a second certificate	Admin	C	`certificates.controller.js:95-181`; `bursary_certificates` UNIQUE `award_id`	`bursaryCertificates.test.js` (12)	V

ID	Requirement (abbreviated statement)	Actor	Pri	Implementation	Verification	Status
FR-23	Revoke a certificate with a reason; public verification then reports `revoked`	Admin	C	`certificates.controller.js:213-249`	`bursaryCertificates.test.js`	V
FR-24	Report `not_current` when the award is no longer approved or its amount changed since issue	System	C	`certificates.controller.js:251-290`	`bursaryCertificates.test.js`	V
FR-25	Dashboard: registry, duplicate, award and payment totals, six-month trend, recent audit activity	Admin	S	`adminDashboard.controller.js`; `pages/AdminDashboard.jsx`	Authorisation and synthetic counts in `roleAuthorization.test.js`; counter arithmetic not individually tested	P
FR-26	Date-range PDF export of registration, disbursement, duplicate and accountability summaries	Admin	S	`reports.controller.js`; `routes/reports.routes.js:8-13`	Range validation tested; PDF content verified by inspection only	P
FR-27	Audit every state-changing operation with actor, action, table, record and details	System	M	`services/auditLog.js`; every controller	`workflow.test.js` #10 (ten-action sequence)	V
FR-28	Read and update the duplicate, high-priority and model cut-offs within 0-1, and audit the change	Admin	S	`settings.controller.js`; `config/settings.js`	Denial tested in `roleAuthorization.test.js`; validation by inspection	P
FR-29	List users and create administrator or field-officer accounts	Admin	S	`users.controller.js`	`roleAuthorization.test.js`; last-admin guard by inspection	P
FR-30	Report liveness and readiness separately; readiness fails closed on a missing table or wrong database identity	Operator	S	`index.js:69-87`; `services/databaseReadiness.js`	`test/databaseReadiness.node.js`; `schemaReadiness.test.js` (5); `health.test.js`	V

6.1 Proposal requirements recorded for completeness

ID	Proposal requirement	Delivered state	Classification
FR-31	Random Forest over four similarity features	XGBoost over six features in production; Random Forest retained only as an unevaluated research artefact	Implemented with a documented design change (G-21)
FR-32	NEMIS / national identifier verification integration	Not implemented; excluded in the proposal	Not implemented, as scoped
FR-33	Offline operation and native mobile applications	Not implemented; responsive web only	Not implemented, as scoped

6.2 Functional gaps recorded as gaps

ID	Gap	Consequence	Evidence
FR-34	No candidate blocking: each registration is compared with the whole registry; the administrator rescan is quadratic	Model calls grow linearly with registry size; review cost grows with N	`students.controller.js:69-73`; `docs/ml-decision-sheet-2026-10-30.md` §6
FR-35	No re-issue path for a student QR token	A row created without a token cannot be given one through the API	`docs/qr-verification-2026-10-30.md` §6
FR-36	No token expiry or rotation	A token remains valid for the life of the student row	`docs/qr-verification-2026-10-30.md` §6
FR-37	Payments are recorded, not executed	No bank, mobile-money or treasury integration	`disbursements.controller.js`

7. Non-functional requirements register (NFR-01 … NFR-21)

ID	Requirement	Measure or constraint implemented	Evidence	Status
NFR-01	Screening completes within three seconds	<= 3 000 ms for a 50-candidate registration; 22.67 ms measured for 50 pairs (2026-10-30)	`tests/performance.test.js`	P: satisfied at 50 candidates; no measurement at registry scale
NFR-02	Password storage resists offline attack	bcrypt, cost factor 12	`auth.controller.js:26`	V as implementation; cost not benchmarked on the host
NFR-03	Session credentials are signed and time-bounded	HMAC JWT, `expiresIn: 8h`, secret >= 32 bytes, placeholder rejected at start-up	`auth.controller.js:55-59`; `config/jwtSecret.js`	V
NFR-04	Authorisation does not trust a client-supplied claim	Role re-read from the `users` row on every request; inactive users refused	`middleware/auth.js:20-24`	V
NFR-05	Identifiers are not exposed by default in the queue	Last-three-character masking plus an audited reveal endpoint	`duplicateFlags.controller.js:37-42,106-107`	V
NFR-06	Public verification discloses no beneficiary data	Student endpoint returns `{found}` only; certificate endpoint returns authenticity facts only	`verification.controller.js`; `certificates.controller.js:53-89`	V
NFR-07	Public endpoints are rate limited	10 requests / IP / 15 min (student); 20 / IP / 15 min (certificate); separate counters	`middleware/verificationRateLimit.js`; `certificateRateLimit.js`; `rateLimitWindow.js`	V in-process only; not shared across instances
NFR-08	Sign-in is rate limited	10 attempts / IP / 15 min	`middleware/loginRateLimit.js`	V
NFR-09	Request bodies are bounded	JSON limit 100 kB	`index.js:64`	V
NFR-10	Errors do not leak internals	5xx returns a generic message; logs carry no stack, message or credential	`middleware/errorHandler.js:59-78`	V
NFR-11	Browser origins restricted by allowlist	Unlisted origin rejected before routing; malformed configuration stops start-up	`config/cors.js`; `index.js:40-47`	V
NFR-12	Integrity enforced by the database, not only the application	CHECK constraints for roles, school list, award decision, verification, payment and receipt status, clearance and acknowledgement completeness, certificate status and token format; foreign keys throughout	`db/schema.sql`; migrations	V
NFR-13	Concurrent registrations do not interleave	Transaction plus `pg_advisory_xact_lock`; candidates selected `FOR UPDATE`	`students.controller.js:67-73`	P: lock exists and is used; no concurrency test exists
NFR-14	Usable by non-technical officers on varied devices	Responsive layout, role navigation, plain-language screening messages, single-line status pages	`components/AppLayout.jsx`; `pages/StudentRegistration.jsx`; `pages/Verification.jsx`	P: browser-checked; no usability study or acceptance session
NFR-15	Maintainable and reproducible	One feature-schema module shared by training and serving; ordered migrations; Jest, native `node --test` and Python `unittest`; pinned requirements	`ml-service/feature_engineering.py`; `db/migrations/`; `ml-service/requirements.txt`	V

ID	Requirement	Measure or constraint implemented	Evidence	Status
NFR-16	Operable within a zero-cost student budget	Container with `minReplicas 0`; free App Service tier; PostgreSQL stopped between demonstrations; no log workspace	`Dockerfile`; `docs/azure-deployment-log-2026-10-31.md` §1-§4	P: the zero-cost configuration is recorded, but no billed month has been reconciled against the budget and the cost model is an owner decision
NFR-17	A human makes every duplicate and payment decision	Scores create cases only; approval, disbursement and issue are role-gated human actions	`duplicateFlags.controller.js:452-503`; `bursaryAwards.controller.js:151-209`	V
NFR-18	Recovery from database loss with verified evidence	Dump, restore into an independent database, readiness and row-digest comparison rehearsed	`docs/database-recovery-rehearsal-2026-10-31.md`	P: local rehearsal; no scheduled backup job or retention policy in the deployment
NFR-19	Compatible with ordinary modern browsers	SPA over HTTPS using `fetch`, `history.replaceState` and router hash handling only	`frontend/src/**`; `frontend/public/web.config`	P: exercised in Chrome only; Firefox, Safari and mobile browsers unrecorded
NFR-20	Predictable degradation when the predictor is unavailable	Registration aborts with 503 instead of registering an unscreened beneficiary	`students.controller.js:74-96`; `services/duplicateCheck.js`	V by inspection and the 503 assertion path
NFR-21	Audit records attributable and complete for state changes	Every controller writes an `audit_logs` row with actor, action, table, record and JSON details	`services/auditLog.js`; `workflow.test.js` #10	P: attributable and complete in every tested workflow; append-only is convention only and is not enforced by the schema

8. Counts and status summary

Group	Count	V	P	N	X
Functional requirements delivered (FR-01 … FR-30)	30	24	6	0	0
Proposal requirements carried as deviations (FR-31 … FR-33)	3	0	0	0	3
Functional gaps reported as gaps (FR-34 … FR-37)	4	0	0	0	4
Non-functional requirements (NFR-01 … NFR-21)	21	14	7	0	0

The six partially verified functional requirements are FR-12 (eligibility arithmetic), FR-19 (not every reason code individually asserted), FR-25 (counter arithmetic), FR-26 (PDF content), FR-28 (cut-off validation) and FR-29 (last-administrator guard); every other delivered functional requirement carries a route-level or workflow-level test. The seven partially verified non-functional requirements are NFR-01, NFR-13, NFR-14, NFR-16, NFR-18, NFR-19 and NFR-21. These counts are obtained by counting the status cells in §6 and §7, not maintained as a separate figure, so a change to a row changes the count.

Two **V** verdicts rest partly on inspection and say so in the row: NFR-02 (bcrypt cost factor is set but not benchmarked on the host) and NFR-20 (the 503 path is asserted and the surrounding branch was read). Every other **V** rests on an executed test or a recorded observation. The **N** column is empty because every implemented requirement has at least a route-level test, which is not a claim that every branch is covered; branch-level coverage is stated in `SBMS_TEST_EVIDENCE_CATALOGUE.md` §7.

9. Traceability entry points

Question	File that answers it
Which endpoint implements a requirement?	`SBMS_API_ENDPOINT_CATALOGUE.md` §3
Which table and column stores the result?	`SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md` §4-§8
Which test proves it, and on what date?	`SBMS_TEST_EVIDENCE_CATALOGUE.md` §3-§6

Question	File that answers it
Which claim is unsupported, and why?	`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §4-§5
Which objective does it serve?	`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §1
Which gap is still open?	`DOCUMENTATION_GAP_REGISTER.md` §1-§3

Statement of provenance. Every row above was read from the revision named in the header, not from an earlier draft. Where the repository contradicts a chapter, the contradiction is recorded in `SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §3 rather than smoothed away here.